

Practical 3

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Date of assignment:	Date of submission:

Q1. Write a program to perform basic operation, indexing and slicing on arrays.

Code:

```
import array
arr=array.array('i',[10,20,30,40,50])
print("Original Array: ",arr)
arr.append(60)
print("After appending 60: ",arr)
arr.insert(2,25)
print("After inserting 20 at index 2:",arr)
arr.remove(40)
print("After removing 20: ",arr)
popped_element=arr.pop()
print("After popping last element(",popped_element,"):",arr)
print("element at index0: ",arr[0])
print("element at index -1(last element):",arr[-1])
print("slice from index 1 to 4: ",arr[1:5])
print("slice with step of 2:",arr[::2])
print("reverse the array using slicing: ",arr[::-1])
```

Output:

```

===== RESTART: C:/Users/DELL/OneDrive/Desktop/Mehek/SEM 3/Python/prac3a.p
Original Array:  array('i', [10, 20, 30, 40, 50])
After appending 60:  array('i', [10, 20, 30, 40, 50, 60])
After inserting 20 at index 2: array('i', [10, 20, 25, 30, 40, 50, 60])
After removing 20:  array('i', [10, 20, 25, 30, 50, 60])
After popping last element( 60 ): array('i', [10, 20, 25, 30, 50])
element at index0:  10
element at index -1(last element): 50
slice from index 1 to 4:  array('i', [20, 25, 30, 50])
slice with step of 2: array('i', [10, 25, 50])
reverse the array using slicing:  array('i', [50, 30, 25, 20, 10])
>>>

```

Q2. Write a program to implement mathematical functions on array.

Code:

```

import numpy as np

arr = np.array([1,4,9,16,25])

print("Original Array: ", arr)
print("Square Root: ", np.sqrt(arr))
print("Square: ", np.square(arr))
print("Cube: ", np.power(arr,3))
print("Log(base e): ", np.log(arr))
print("Exponential (e^x):", np.exp(arr))
print("Sine:", np.sin(arr))
print("Cosine:", np.cos(arr))
print("Maximum value: ", np.max(arr))
print("Minimum value: ", np.min(arr))
print("Sum of elements: ", np.sum(arr))
print("Mean (Average): ", np.mean(arr))

```

Output:

```

===== RESTART: C:/Users/DELL/OneDrive/Desktop/Mehek/SEM 3/Python/prac3a.py =====
Original Array: [ 1  4  9 16 25]
Square Root: [1. 2. 3. 4. 5.]
Square: [ 1 16 81 256 625]
Cube: [ 1 64 729 4096 15625]
Log(base e): [0. 1.38629436 2.19722458 2.77258872 3.21887582]
Exponential (e^x): [2.71828183e+00 5.45981500e+01 8.10308393e+03 8.88611052e+06
7.20048993e+10]
Sine: [ 0.84147098 -0.7568025  0.41211849 -0.28790332 -0.13235175]
Cosine: [ 0.54030231 -0.65364362 -0.91113026 -0.95765948  0.99120281]
Maximum value: 25
Minimum value: 1
Sum of elements: 55
Mean (Average): 11.0

```

Q3. Write a program to perform array aliasing and copying.

Code:

```
import numpy as np
```

```
print("----- Array Aliasing using Numpy -----")
```

```
arr1 = np.array([11, 22, 33, 44])
```

```
arr2 = arr1
```

```
print("Original arr1:", arr1)
```

```
print("Aliased arr2:", arr2)
```

```
# Modify arr2
```

```
arr2[1] = 99
```

```
print("After Modifying arr2[1] = 99")
```

```
print("arr1 (also changed):", arr1)
```

```
print("arr2:", arr2)
```

2. Copying Example Code

```
print("\n----- Array Copying Using Numpy -----")
```

```

arr3 = np.array([51, 52, 53, 54])
arr4 = arr3.copy()  # independent copy

print("Original arr3:", arr3)
print("Copied arr4:", arr4)

# Modify arr4
arr4[2] = 100

print("After modifying arr4[2] = 100")
print("arr3 (unchanged):", arr3)
print("arr4:", arr4)

```

Output:

```

===== RESTART: C:/Users/DELL/OneDrive/Desktop/Mehe
----- Array Aliasing using Numpy -----
Original arr1: [11 22 33 44]
Aliased arr2: [11 22 33 44]
After Modifying arr2[1] = 99
arr1 (also changed): [11 99 33 44]
arr2: [11 99 33 44]

----- Array Copying Using Numpy -----
Original arr3: [51 52 53 54]
Copied arr4: [51 52 53 54]
After modifying arr4[2] = 100
arr3 (unchanged): [51 52 53 54]
arr4: [ 51  52 100  54]
>>

```